PY-STEM · ACTIVITY PYS-10 · OPTICS + LIGHT

Spectra Sleuth Showdown

Can two white lights hide different color fingerprints?

Win condition: Most correct spectrum matches and strongest safe fireworks-science explanation.

THE SCIENCE YOU ARE USING

Diffraction splits light. A diffraction grating bends different colors by different amounts, spreading light into a spectrum. Two lights that look the same white can have very different spectra: a smooth rainbow from a hot filament, a broad colored band from a white LED, or separated bright lines from a neon or gas source.

Spectra as fingerprints. Each light source has a characteristic spectrum, its fingerprint. This is the same science behind firework colors, where different metal salts emit specific colors. You match mystery sources to clue cards using their spectra, no flame needed.

HOW TO RUN IT

01 **Look through the grating.** View a white light through the grating and watch it spread into colors.

02 **Compare two sources.** Look at a white LED and a neon light. The LED shows a broad band; the neon shows separate bright lines.

03 **Sketch the spectra.** Draw what you see for each mystery source on the spectrum sketch sheet: continuous band or separate lines.

04 **Match to clue cards.** Use your sketches to match each source to its spectrum card.

05 **Connect to fireworks.** Explain how this links to the colors in a firework display, safely.

Safety: No flame tests, lasers, or direct sun viewing. Diffraction glasses are not eye protection. Only staff handle plugged-in or hot light sources. Allergy: None.

MATERIALS

Diffraction glasses or gratings	per student
White LED and neon or gas sources	shared
Incandescent source	if available
Spectrum cards	set
Museum clue sheets	per team

YOUR PLAN AND DATA

Observe and sketch each mystery spectrum before matching it to a source card.

Mystery source or card number and observed pattern:

Proposed source match and visible evidence:

Second mystery source or card and observed pattern:

Proposed source match and visible evidence:

DEFEND YOUR MATCHES

Which visible feature best separated the source types?

How does a line spectrum connect to fireworks colors?

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Correct matches	40
Spectrum sketches	20
Explanation	20
Exhibit connection	20
Total	100